



Wind Energy Assessment

Phase 2- Mappings & Notes

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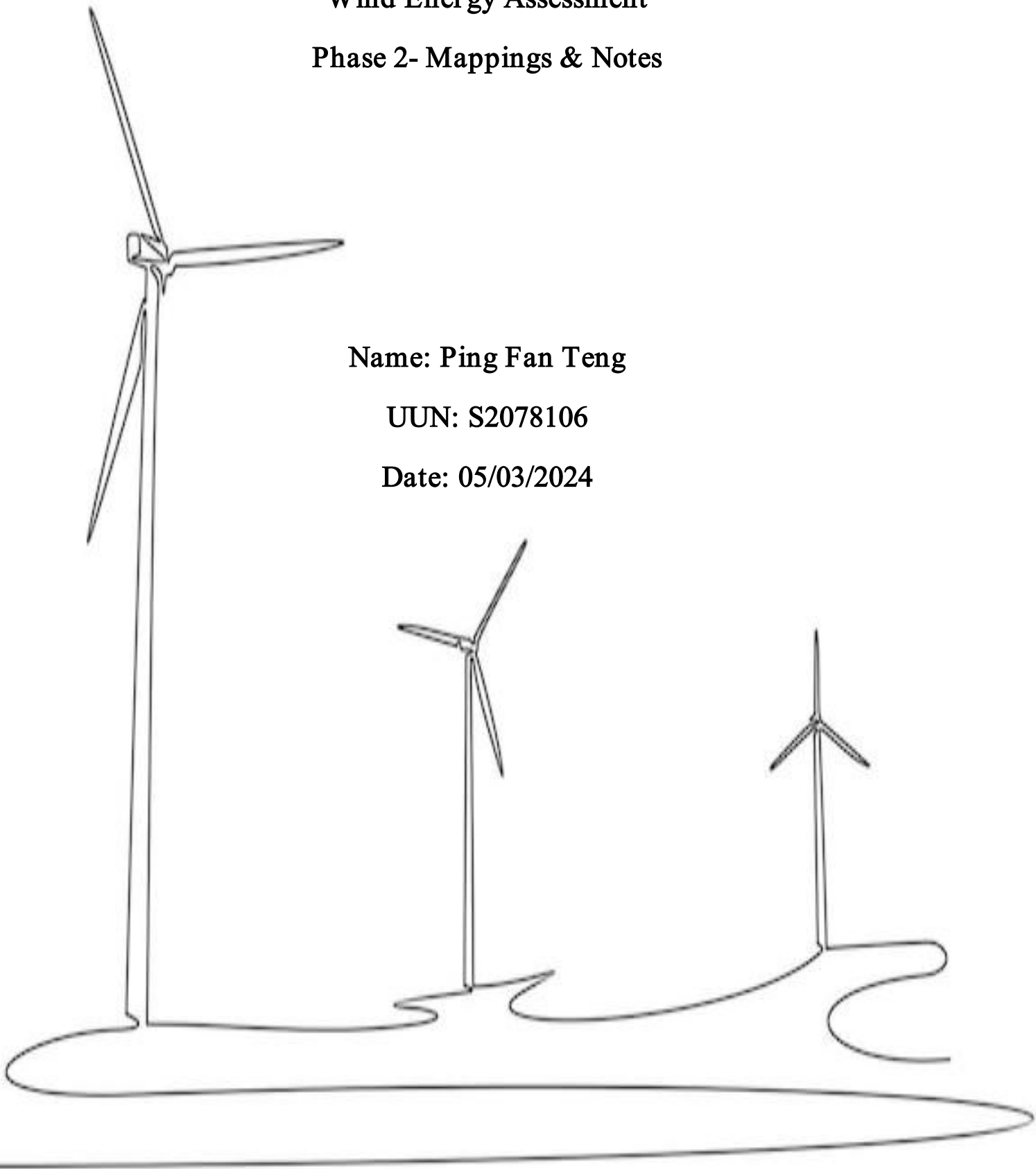




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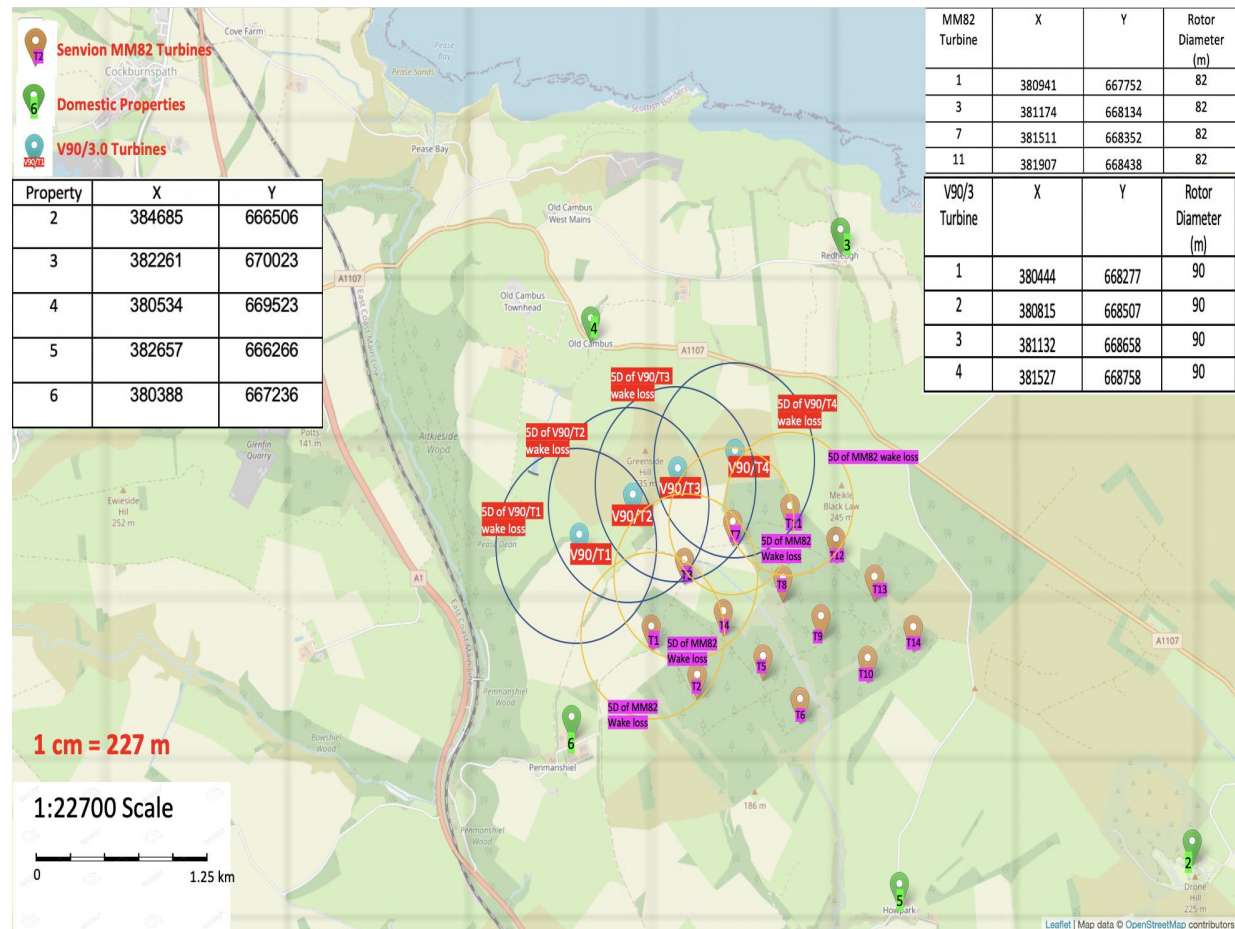
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V90/3.0 Location and Wake Loss



Map 1. Wake loss and set back distance between new and old turbines

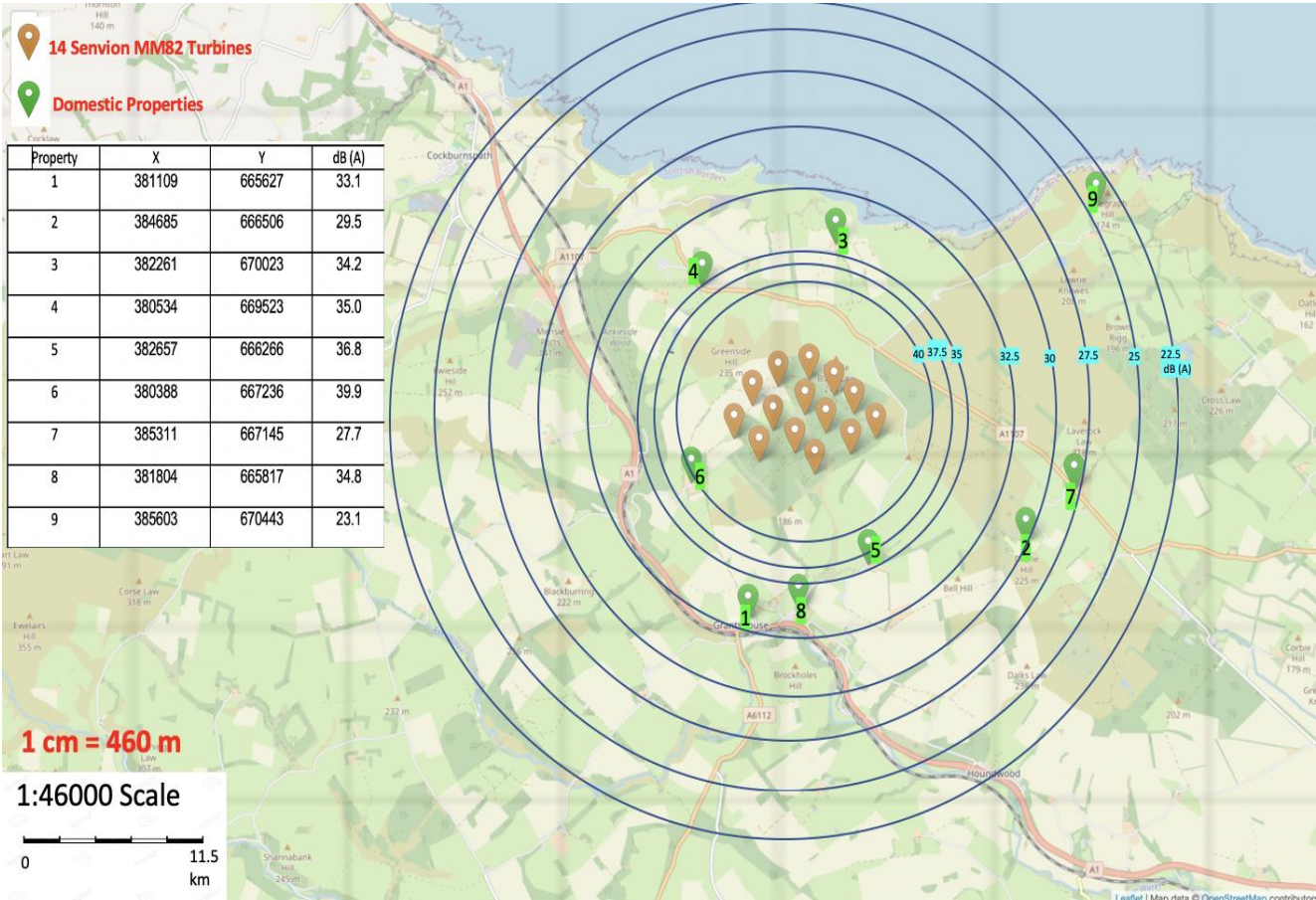
Based on Excel spreadsheet calculations, it was decided to select four Vestas V90 3 MW turbines for installation at the selected site shown above.

From the wind characteristics of Penmanshiel, it was established that the prevailing wind direction originates from the SW and WSW. Therefore, it would be less productive to place new turbines in the north-eastern section of the existing wind farm, where the prevailing wind would be deflected or blocked by the existing turbines. In contrast, placing new turbines to the south-east of the wind farm would partially obstruct wind flow to the existing turbines. A more suitable area was identified in the north-western section of the wind farm, with a 10D setback distance from the nearest residential properties.

The Jensen–Katic model [1] was used to estimate the percentage reduction in wind velocity. The results were found to be insignificant, given that the dominant wind direction is from the WSW and SW, and the maximum spacing between newly installed turbines is approximately 3D of the rotor diameter, with an additional spacing of approximately 5D between the new and existing turbines.

In addition, the selected site may extend beyond the current boundary of the Penmanshiel wind farm; however, this can be addressed through negotiation with nearby landowners.

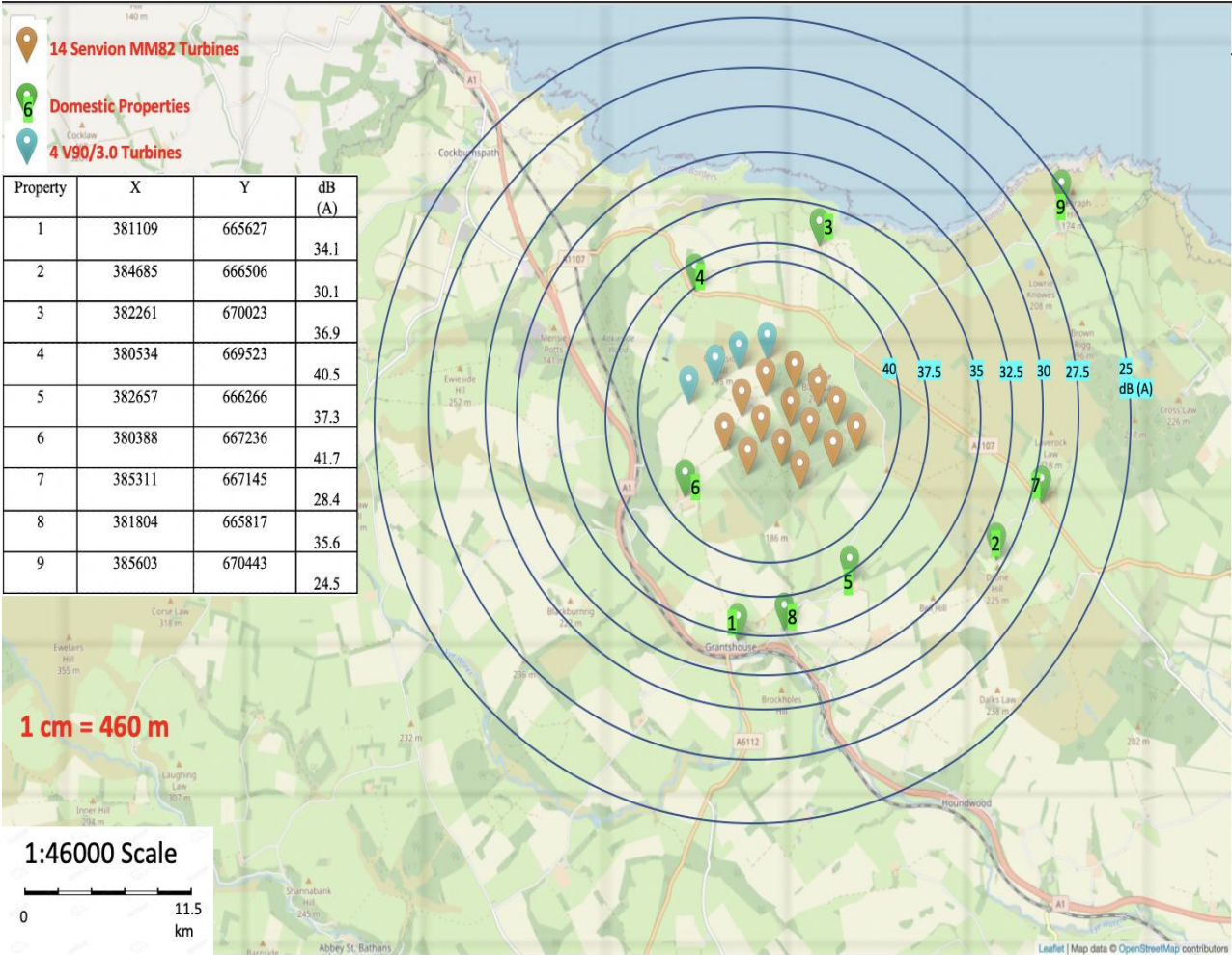
Noise Assesment – Current turbines



Map 2. The total perceived noise from the existing MM82 turbines at Penmanshiel wind farm.

Perceived noise levels at the Penmanshiel Wind Farm are within the range of 35–40 dB(A), indicating that the previous noise assessment for the wind farm remains valid. Refer to the Excel spreadsheet of the noise calculations for further details.

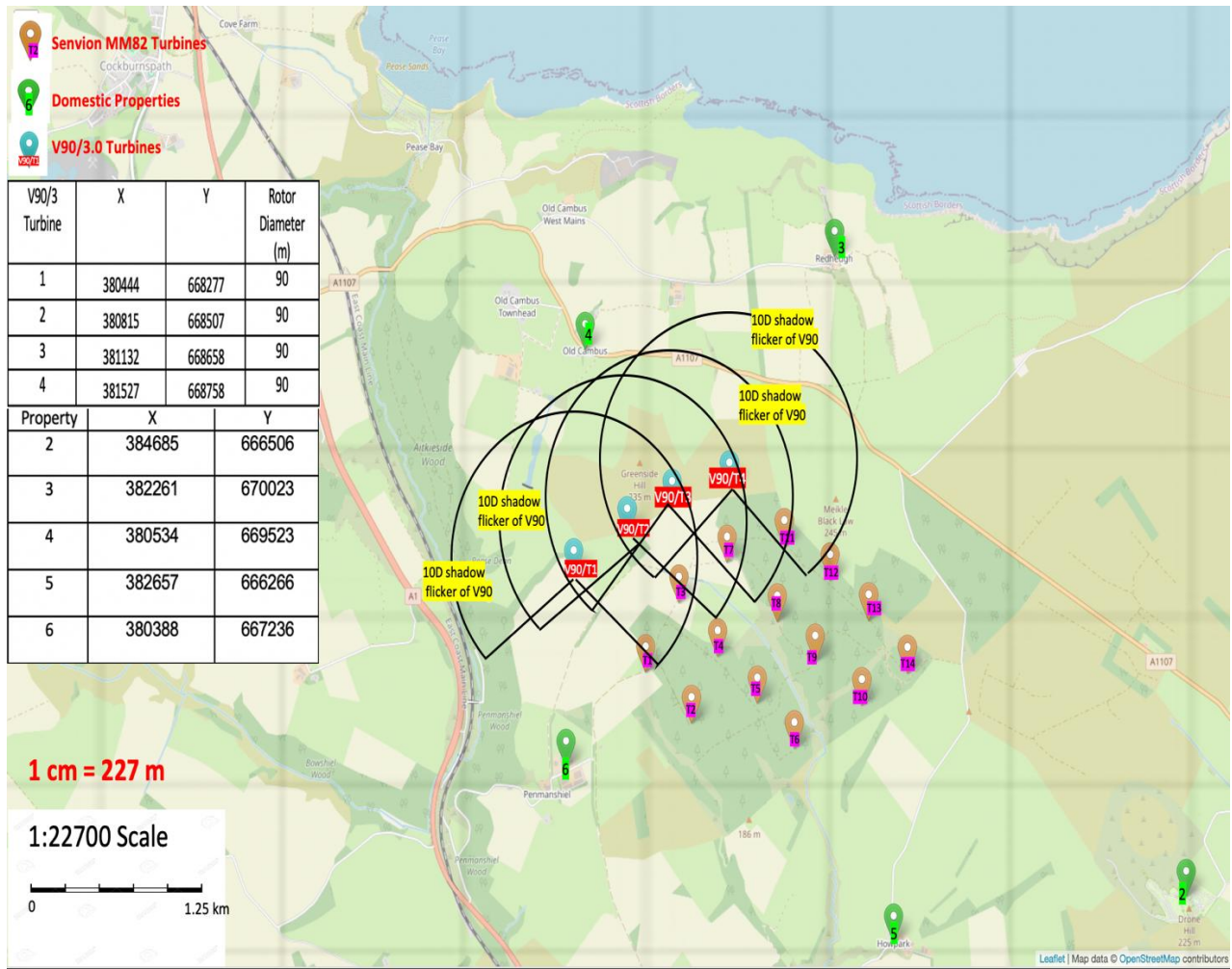
Noise Assessment- New installed turbines



Map 3. The total perceived noise from the existing turbines with new installed turbines

Closest properties, such as Penmanshiel Farm and Old Cambus, may be affected by the changes. However, based on the averaged background noise limit from previous planning documents [2], it was found that the new total perceived noise remains within the averaged noise limit and is offset by +5 dB(A). Therefore, it is not considered a major issue. If complaints arise, it is suggested to install noise barriers around Penmanshiel Farm and Old Cambus. Otherwise, it is recommended to replace the windows with more soundproof alternatives [3].

Shadow Flicker



Based on UK Planning Guidelines [4], all nearby residential properties are located more than 10 rotor diameters from the locations of the four newly installed turbines; therefore, it is reasonable to state that no shadow flicker mitigation is required. ($10D = 10 \times 90 = 900$ m)

In addition, previous planning documents [2] confirm that shadow flicker from the existing turbines at Penmanshiel Wind Farm has already been assessed, and a similar assessment applies to the newly installed turbines.

Ice throw

Based on $d = 1.5(D + H)$, where d is the maximum throwing distance (m), D is the rotor diameter (m), and H is the hub height (m), it is calculated that the maximum throwing distance for the V90/3.0 is 270 m. This distance has no impact on the nearest residential properties or any nearby infrastructure. Similarly, as stated in the previous planning statement [2], the effects of ice throw were considered insignificant. Therefore, it is concluded that there is no need to consider any adverse impacts arising from ice throw associated with the installed turbines.

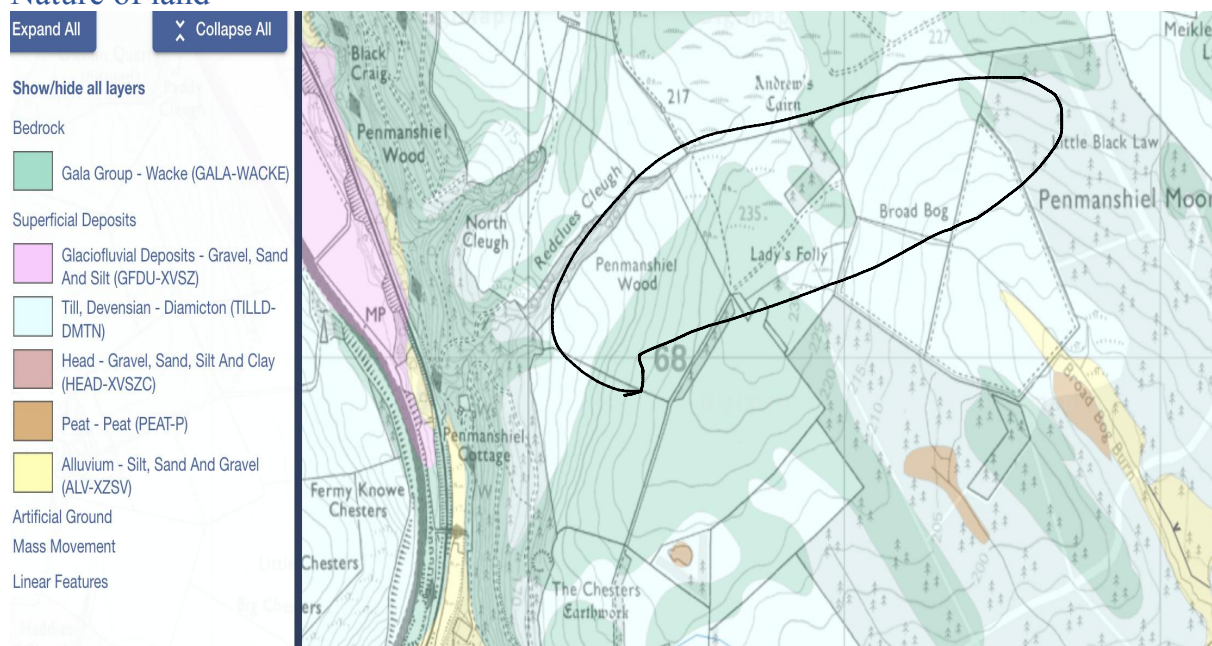
Electromagnetic Interference

Reviewing the previous planning statement [2], it was noted that television reception in several homes was affected by the Penmanshiel Wind Farm in 2011. It was further stated that most households had switched to alternative solutions to restore their television reception, and that any applicants affected by issues caused by the Penmanshiel Wind Farm were expected to take their own action to resolve them.

Further investigation will be undertaken if complaints are raised by nearby homes regarding television reception issues. However, at present, there is generally no need to consider electromagnetic interference effects caused by the newly installed wind turbines.

Environmental/Social Impact

Nature of land



Map 5. Geographical map of the site from Digimap

The newly located turbines are situated on sandstone; the dry ground provides good structural stability for the wind turbines and complies with planning document [2], which states that marshy or wet sand areas should be avoided. In addition, construction activities will avoid underground water and gas pipelines.

However, this may have a slight impact on the nature of the bedrock, although this has been assessed and it is not identified by the Scottish Government as a protected natural feature.

With regard to agricultural development, there is no evidence of farming operations at the selected site.

Studies have shown that wind farms may cause a slight increase in soil and near-surface air temperature during the night [5]. However, this has been quantified at only 0.18 °C, which may

increase plant respiration and thus carbon dioxide release during nighttime. Nevertheless, the Penmanshiel area consists primarily of short grassland, and the soil conditions at the selected site are not suitable for plant growth.

The installation of the new wind turbines has taken into consideration the need to avoid any nearby watercourse crossings.

Historical/Cultural Affects

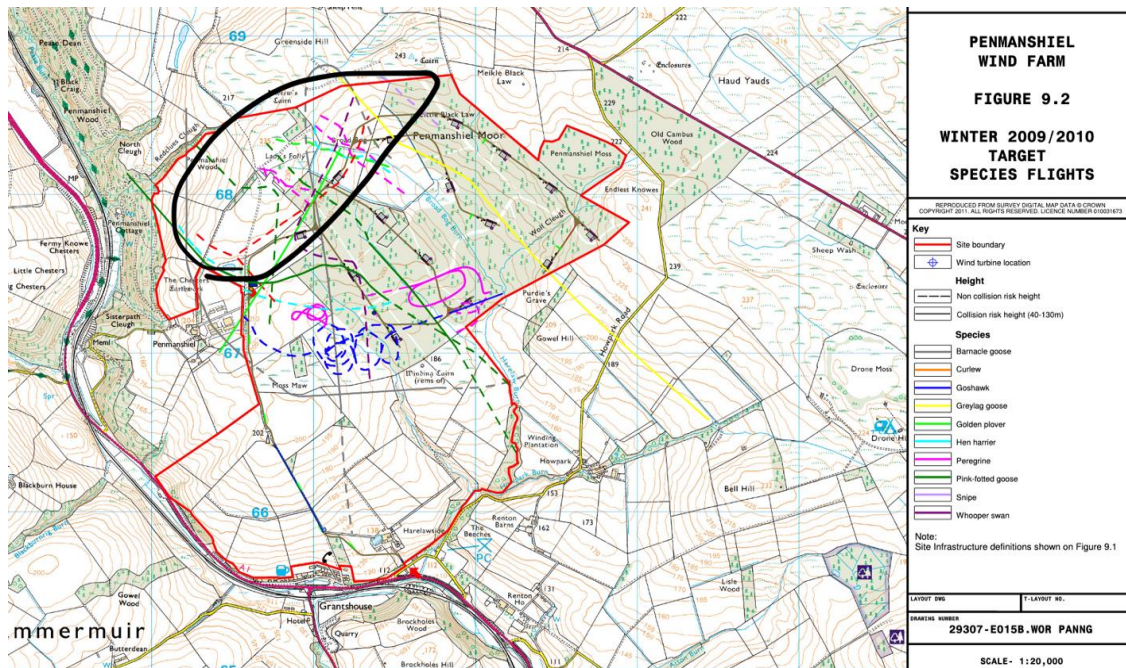
Table 10.10: Predicted direct effects on cultural heritage features within the proposed wind farm site

Site no	Site name	Sensitivity of receptor	Predicted effect	Magnitude of effect	Significance of effect
6	Penmanshiel, cairn(s)	Low	Direct: part of recorded extent of site is within shelter belt. Any tree felling and tree stump removal may disturb buried remains	Low	Negligible
7	'Lady's Folly' Penmanshiel, cairn	Low	Direct: recorded location of site is within shelter belt. Any tree felling and tree stump removal may disturb buried remains	Medium	Minor
11a/b	'Lady's Folly' Penmanshiel, barrow, cairn	Low	Direct: In close vicinity to shelter belt. The exact extent and location of buried remains cannot be accurately predicted. Any tree felling and tree stump removal may disturb buried remains.	Medium	Minor
16	Moss Maw, Penmanshiel, cairn(s), knife (flint) and polished axehead (flint)	Low	Direct: part of recorded extent of site would be crossed by proposed new access track. Possible disturbance of buried remains.	Low	(Negligible)
24	Field boundaries	Negligible	Direct: standing drystone field walls crossed by proposed access tracks	Imperceptible	Negligible

From the past planning assessment of historical and cultural impacts of wind turbines at the Penmanshiel Wind Farm [6], it is reasonable to state that the selected site has minimal impact on heritage buildings and remains. The proposed site is also located close to Lady's Folly [6].

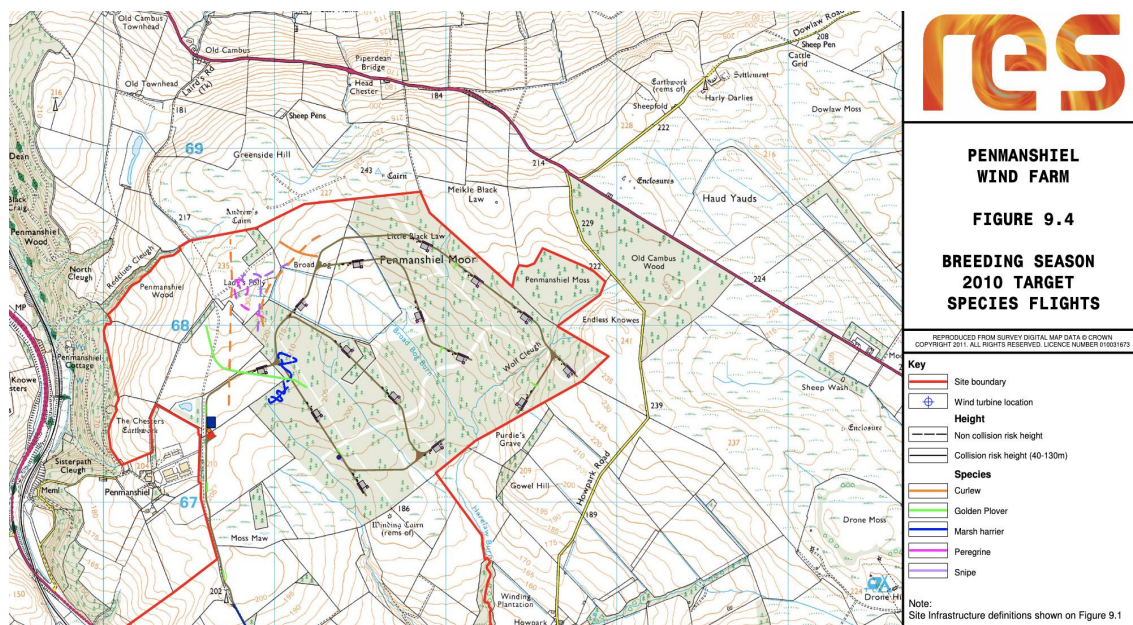
This assessment was carried out across several locations that could experience indirect effects, such as the Walled Garden, Lowries Knowes, Ewieside Hill, Harly Darlies Iron Age Settlement, and the Atton Settlement [6]. In addition, direct visual impacts were already assessed in this document; therefore, no significant issues were identified in this regard.

Bird Migrations



Map 6. Map from planning document [2]

From the previous planning document [2], it is acknowledged that the selected area for the new wind turbines has little or negligible impact on different bird migration routes. The map above illustrates that the area circled in black has a low impact on bird species, as no collision risk is expected.



Map 7. Map from planning document [2]

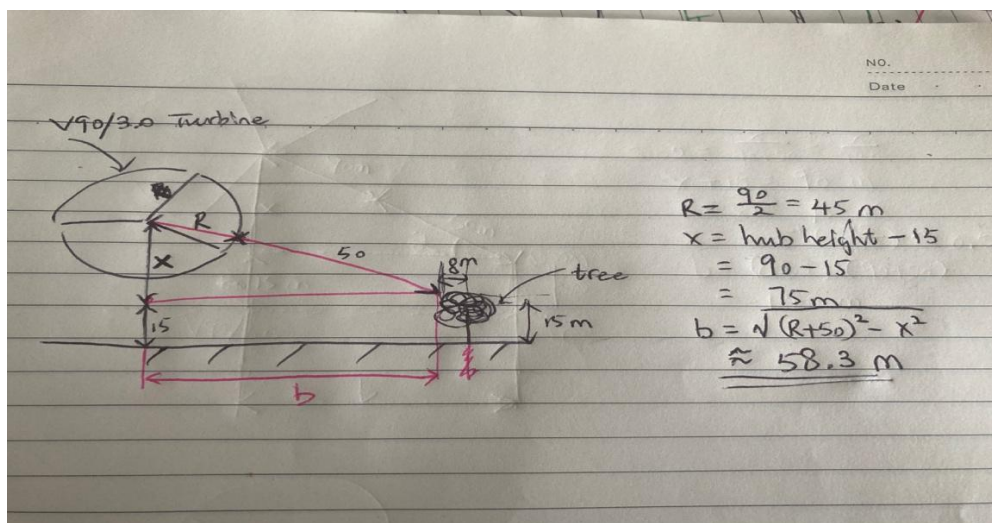
From past records, during the breeding season, the selected site for the newly installed turbines might affect the golden plover. However, this can be disregarded as the survey was carried out more than 10 years ago; it is likely that the birds have since found other suitable nesting areas. Hence, the risk of affecting the birds is quite unlikely when constructing the new turbines at this site.



Pic 1. A golden plover

Bats Issues

Based on the criteria Natural England TIN051 [7], I have calculated the safety clearance distance of b to be equal to 58.3m for the selected V90/3.0 wind turbines.



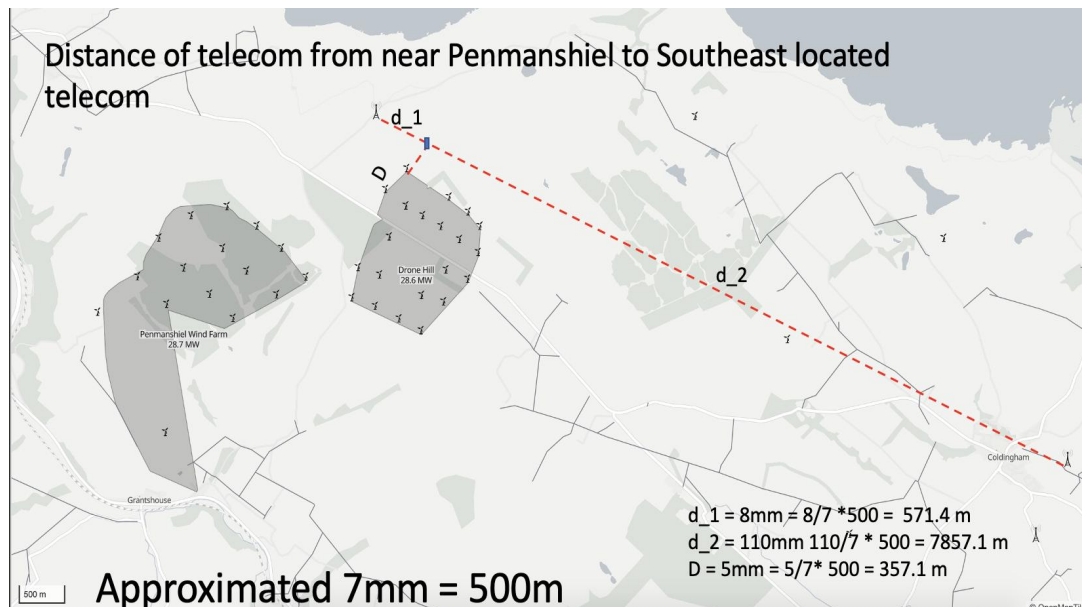
Pic 2. Calculations for bat clearance distance

Taking that into consideration, I have avoided placing turbines within a 58.3 m range of the nearest tree area. For illustrative purposes, please refer to Map 8; note that this map is not to scale and was generated using Digimap.



Map 8. Aerial view of the 4 V90/3.0 turbines (map is not to scale)

Telecom issues (For Drone Hill Wind Farm)



Map 9. Open infrastructure map of the telecoms

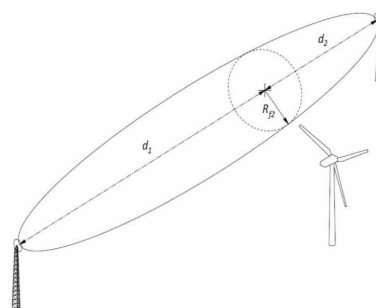


Figure 21 Model showing the 2nd Fresnel radius R_{f2}

$$R_{f2} = \sqrt{\frac{600d_1d_2}{f(d_1+d_2)}}, \text{ where } d_1 \text{ and } d_2 \text{ are in km, frequency } f \text{ in } 10^9 \text{ Hz, } R_{f2} \text{ in meters.}$$

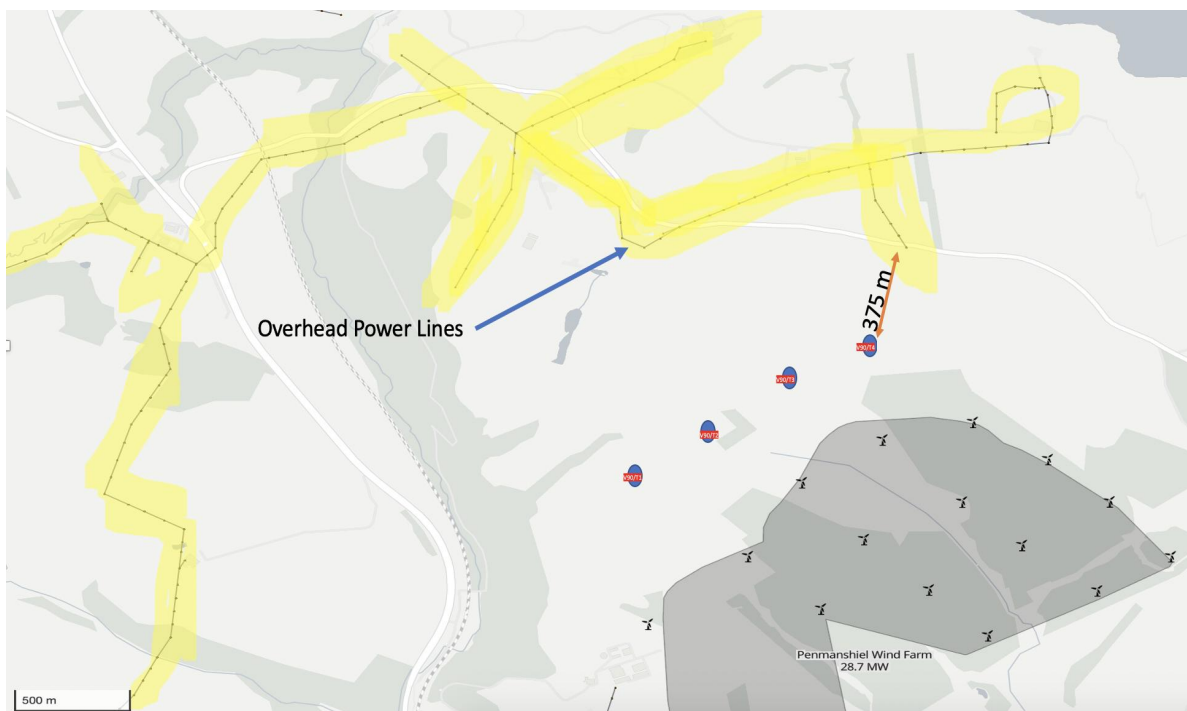
assuming $f = 1.5$ GHz (UK averaged frequency band of cell towers)

$$\text{Then } R_{f2} = \sqrt{\frac{600d_1d_2}{f(d_1+d_2)}} = \sqrt{\frac{600 \times 0.5714 \times 7.857}{1.5 \times (0.571 + 7.857)}} = 25.3 \text{ m}$$

Just for reference that the Drone Hill Wind Farm has zero impact on the closest telecom.

Based on the new locations site of the turbines, there is a +250 m blade tip clearance to the telecommunication links. This is because there is only 1 Telecom in the northern part away of +2km. Hence it is also complied with the planning documents [2].

Overhead Cables/Power Line



Map 10. Overhead power line layout and V90/3.0 turbine location

The distance between the newly installed turbines and the nearest overhead power line is approximately 375 m. This is greater than 1.1 times the turbine tip height, which is calculated as $1.1 \times (90/2 + 90) = 148.5$ m.

Therefore, the selected site for the new turbines has no impact on the overhead power line. In addition, this proximity may be beneficial in the future if the installed wind turbines require connection to a closer grid via the overhead power line.

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